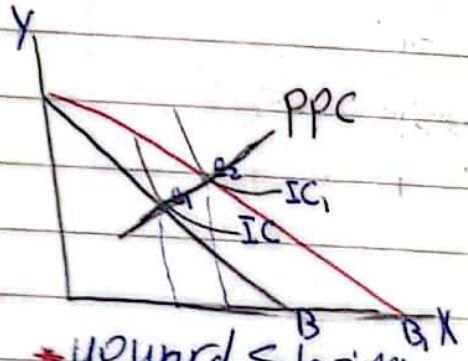
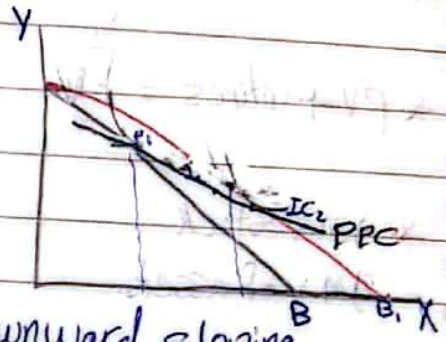
⇒ show graphically PPC, incase of:

inelastic Dx
⇓
% ΔP > % ΔQd



* upward sloping
* $P_x \downarrow \rightarrow Q_x \uparrow, Q_y \uparrow$

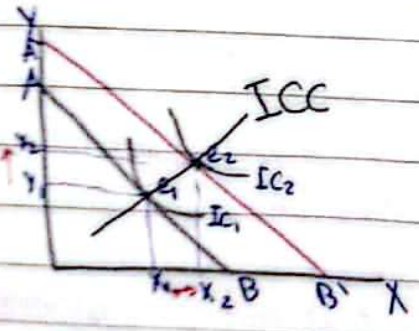
elastic Dx
⇓
% ΔP < % ΔQD



* downward sloping.
* $P_x \downarrow \rightarrow Q_x \uparrow, Q_y \downarrow$

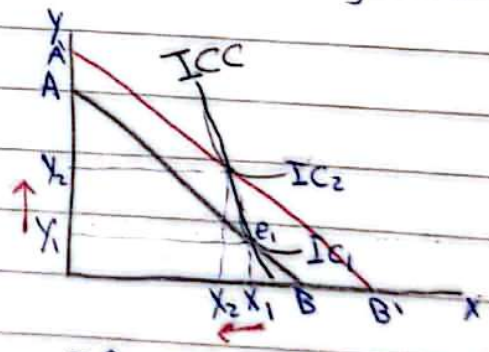
⇒ ICC: show the effect of ΔI on the consumption of goods.
⇒ show graphically the ICC ΔI (I↑) incase of

Normal good (X, Y)



* I↑ → BL shift outwards
* $Q_x \uparrow, Q_y \uparrow$
* upward sloping.

inferior good x



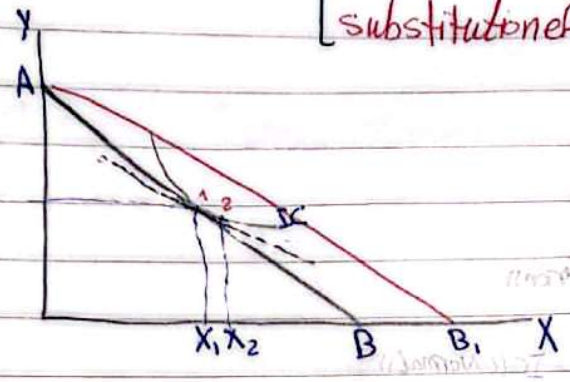
* I↑ BL shift outwards
* $Q_x \downarrow, Q_y \uparrow$
* Backward sloping

PAGE
DATE

⟹ How to differenciate between substitution effect and income effect when $P_x \downarrow$?

[substitution effect] as $P_x \downarrow \rightarrow Q_x \uparrow$

buy more x and less y

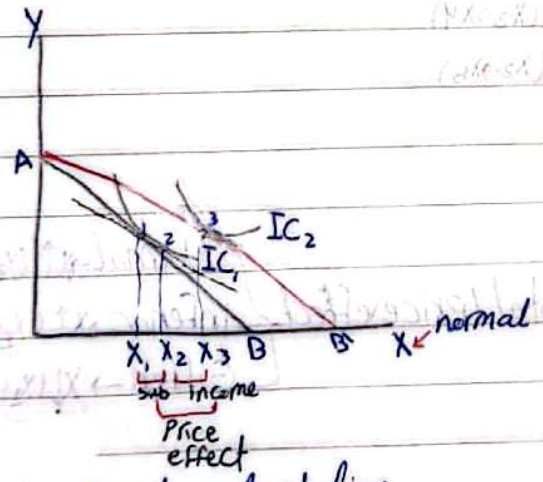


باخذال BL الجديد بصوره
موازيه عشان يلمس ال IC
ويطون لحه اثر ال sub بس

* $P_x \downarrow \longrightarrow$ BL outward rotation
* sub effect → represented by movement along the original IC (1 to 2)
* utility is constant «because we move along the same -old- IC»

[income effect] as $P_x \downarrow$ → purchasing power ↑

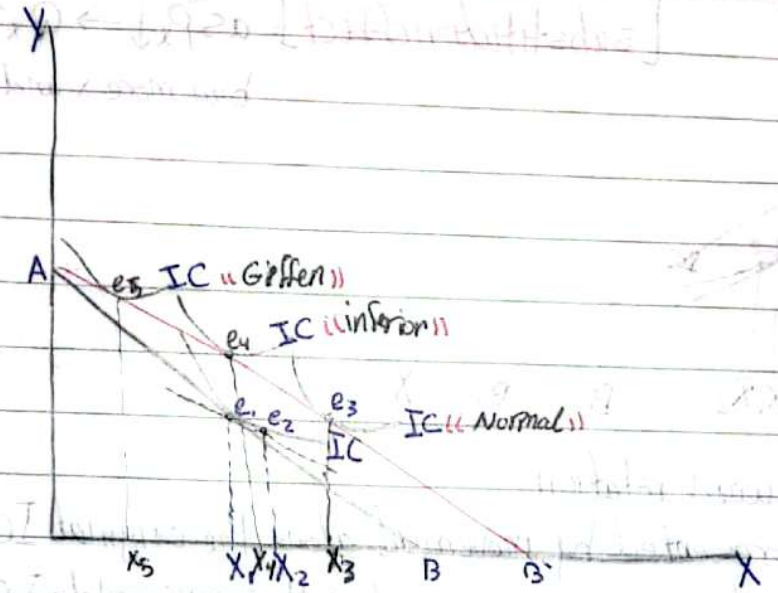
ΔQ_d → normal ↑ / inferior ↓



* $P_x \downarrow$ → B1 outward rotation.
* income effect → represented by movement from lower IC. to higher IC_2 2 to 3
* utility increased «because we move from lower IC to higher IC»

PAGE
DATE

⇒ Differentiate between substitution effect and income effect in case of normal, inferior, and Giffen good X.



* $P_x \downarrow \rightarrow$ BL outward rotation.
* Sub effect (X_1 to X_2) Movement from (e_1 to e_2) along IC_1
*
income effect
- Normal: $e_2 \rightarrow e_3$, $X \uparrow$ ($X_2 \rightarrow X_3$)
- inferior: $e_2 \rightarrow e_4$, $X \downarrow$ ($X_2 \rightarrow X_4$)
- Giffen: $e_2 \rightarrow e_5$, $X \downarrow$ ($X_2 \rightarrow X_5$)

type of good	sub effect	income effect	overall effect
Normal good	$Q_x \uparrow$	$Q_x \uparrow$	$Q_x \uparrow$ D_x downward sloping
inferior good	$Q_x \uparrow$ >	$Q_x \downarrow$	$Q_x \uparrow$ D_x downward sloping
Giffen good	$Q_x \uparrow$ <	$Q_x \downarrow$	$Q_x \downarrow$ D_x upward sloping

* total price effect
- Normal $\rightarrow X \uparrow$ (X_2 to X_3)
- inferior $\rightarrow X \uparrow$ (X_2 to X_4)
- Giffen $\rightarrow X \downarrow$ (X_2 to X_5)